

Ontology-Based Knowledge Graphs for Detecting Advanced Persistent Threats (APTs) with Neurosymbolic AI

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Abstract—APTs are usually well-crafted multi-staged attacks that are low-signal and context dependant therefore they can succeed in avoiding traditional intrusion detection system. In this paper, we propose a novel neurosymbolic AI framework that is composed of ontology-based KG and GNN to enhance the detection of APTs in enterprise networks. We create a cyber security ontology that represents APT tactics, techniques and procedures (TTPs) using notions from MITRE ATT&CK and the Cyber Kill Chain, and we include in this representation explicit causal and temporal relationships between entities such as malicious infrastructure, hosts, processes, attack phases. Variety: Heterogeneous sources of data such as network logs and open-source threat intelligence (e.g., STIX/TAXII) are fed into a dynamic KG which facilitates context-aware reasoning. A heterogeneous GNN is further used to propagate node and edge level latent representations within this KG, but the symbolic rules learned from the ontology are leveraged to better refine and re-weight neural anomaly scores across plausible attack evolutions (e.g., “reconnaissance → C2 communication”). This neurosymbolic fusion aims to solve the challenges over APT detection, i.e., scarcity of labeled data for zero-day attackers and no interpretability in black-box models. Our method obtains in both DARPA TRACE and MITRE ATT&CK Evaluations the higher precision, recall, F1-score compared to baselines (more than XGBoost and LSTM-based IDSs) generating interpretable alerts thanks to cause-and-effects chains. The framework is capable of near real-time operation and can be operationalised in security operations centres using widely-used large-scale graph analytics libraries such as Neo4j and PyTorch Geometric, allowing users to scale out explainable APT detection as well as proactive threat hunting.

Index Terms—Advanced Persistent Threats (APTs), Intrusion Detection Systems (IDS), Neurosymbolic AI, Knowledge Graphs, Graph Neural Networks (GNNs), Ontology, Threat Intelligence, MITRE ATT&CK, Explainable AI (XAI), Zero-Day Detection

I. INTRODUCTION

APTs are one of the most serious issues of modern cybersecurity. Contrary to opportunistic attacks, APTs are multi-staged, stealthy, but customized for given: targets, institutions,

systems and environments and frequently continue for long time periods all while they are: exfiltrating sensitive data, or disrupting key services [4]. Being low-signal and context dependent, if the subtle lateral movement or command-and-control (C2) like cues are not explicitly defined in the form of signatures, these might easily avoid detection from a traditional IDS or a mere data-driven anomaly activity [12]. Traditional machine learning approaches tend to not generalize well in such a situation, particularly against zero-day adversaries introducing new exploits with minimal or no labeled data.

One promising way of dealing with these challenges is to combine structured domain knowledge and data-driven learning. The knowledge graphs (KGs) based on ontologies (i.e., ontology-based KGs) offer a rich formalism to capture the entities, relationships and their temporal order in complex attack campaigns [9]. With the use of popular constructs such as MITRE ATT&CK [14] and the Cyber Kill Chain [8], TTPs can be described in a structured format, which captures causal and sequential relationships between stages of the attack. Semantic models have the potential to enhance contextual meanings and to be able to reason over incomplete or noisy information.

Heterogeneous, graph-structured data such as enterprise logs and threat intelligence feeds can now be leveraged effectively to learn latent patterns thanks to recent developments in graph neural networks (GNNs) [18]. However, GNNs suffer from the requirement of sufficient labeled data and the lack of interpretability in their learned representations, leading to a lack of trust in sensitive application environments. Neuro-symbolic AI provides a solution to this issue by reconciling the statistical power of learning of neural network approaches – such as GNNs – with symbolic reasoning provided by ontologies and logic rules [2]. Such a hybrid approach combines accurate anomaly detection and transparent descriptions of detected

threats.

In this paper, we present a neurosymbolic solution for APT detection, by ensembles of ontology-based knowledge graphs and graph neural networks. Both network logs and open-source threat intelligence (e.g., STIX/TAXII feeds) are automatically synthesized into a domain-specific KG to reason on dynamic attack graphs. GNNs are used to learn the embeddings of nodes and edges by taking advantage of the hidden relationships, and symbolic rules rule-based reasoning derived from ontological constraints are leveraged in the inference process to detect anomalous behaviors. This design enables the system to emphasize causal attack chains (e.g., “reconnaissance → lateral movement → exfiltration”) that are actionable and interpretable for security analysts. We evaluate our method on standard datasets such as DARPA TRACE and MITRE ATTCK evaluations and show its outperformance against state-of-the-art baselines such as based XGBoost and LSTM-based IDS models. Our contributions can be summarized as follows:

- **Ontology-based APT model:** We develop a domain-specific ontology encapsulating the concepts of MITRE ATTCK and Cyber Kill Chain (CKC) factors for modeling tactics, techniques, and causal sequences of an attack in a dynamic knowledge graph.
- **Neurosymbolic integration:** We design an hybrid decision network which integrates neo-symbolic reasoning and GNN-based representation learning to coordinate the trade-offs among accuracy, robustness and interpretability.
- **Holistic validation:** We evaluate our approach on real-world enterprise datasets and show better performance against state-of-the-art machine learning baselines that do not offer explainable causal alerts.

By linking symbolic and neural models, this .—5—frame-work provides a scalable and understandable means to detect APTs, which has profound implication for the real-time defense and the proactive hunting of threats inside security operations centers.

II. RELATED WORK

Intrusion detection has been an early area of research in cyber security with the initial systems mostly based on signature-based or statistical anomaly detection to capture malicious activities. Although successful against known threats, these measures cannot cope up with Advanced Persistent Threats (APTs) that are stealthy, multi-stage and also context-dependent. To enhance the adaptability, machine learning techniques, such as Support Vector Machines (SVM), Decision Trees, and ensemble models are devised, but most of these models on the current datasets do not generalize well on advancing attack scenarios and struggle with identifying low-signal behaviors of APT campaigns otherwise [4], [6], [12], [13]. In order to detect sequential attack behavior, deep learning-based approaches such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) have also been explored [10, 30], which achieved significant accuracy improvements by leveraging temporal information and spatial patterns. But these methods often require a large amount of

data to work with, are susceptible to zero-day attacks, and tend to have little interpretability, which makes them not suitable for security-sensitive environments where analysts need to reason about the alerts in a transparent manner.

In order to address these limitations, ontology-driven and knowledge graph (KG) based methods have been getting a lot of attention due to their power in representing adversary TTPs in a structured and understandable manner. Such ontologies as MITRE ATTCK and the Cyber Kill Chain are established frameworks for characterizing attacker tactics and techniques, and the evolution of campaigns over time, that have been well-received by both the research community and industry [8], [14]. Based on these frameworks, some works have generated cybersecurity KGs that consolidate different types of information sources, including logs, alerts and open source threat intelligence into a single representation [9], [17]. These systems show promise for contextual analytics across disparate data, while also enhancing the mapping of seemingly unrelated events and their correlation to advanced attack campaigns. However, ontology-based solutions are commonly static and inadaptive to new unseen threats, as they only depend on pre-defined knowledge and do not discover and extract latent behavior patterns, while they could result incomplete if the used ontology is not able to cover new attack tactics.

Meanwhile, graph neural networks (GNNs) have become a new dominating network paradigm to learn from relational and graph-structured data, and thus they are a natural choice for cybersecurity techniques where entities like hosts, processes, events construct inter-connected structures. Intrusion detection and malware analysis attack campaign modeling GNN-based methodologies have been used to a variety of cyber security tasks such as intrusion detection, malware analysis, and attack campaign modeling, achieving superior performance than featurebased classifiers, by capturing the complex dependencies via message passing on nodes and edges [15], [16], [18]. GNNs have been successful in learning embeddings and capturing anomalies, however, they demand plenty of labeled training instances and they usually function as black box model. This lack of interpretability prevents their realistic deployment in security operations centers (SOCs) where interpretable alerts are essential for timely and confident decision making. In addition, modelcentric neural models may overfit to the others dataset, making it hard to defend adaptive attacks.

Neurosymbolic AI has recently been suggested as a promising approach to balance between statistical learning and interpretability. Neuro-symbolic methods, which combine symbolic reasoning with data-driven learning by taking advantage of the pattern recognition ability of neural networks [2], [5], were created to bridge the gap between data-driven learning and explicit knowledge representation. These ways of doing have been prominent in domains such as natural language understanding, vision, and knowledge graph reasoning, providing better performance in terms of both accuracy and interpretability. In the realm of cybersecurity, early attempts such as CyberKG demonstrate the potential of integrating

ontological reasoning with machine learning in threat intelligence and situational awareness [1], [7]. Nevertheless, neurosymbolic methods have not been thoroughly applied for the APT detection. Studies in literature often deal with ontology-based reasoning and machine learning separately as components but do not provide an integrated framework, leading to systems with stronger emphasis on semantic nature but with less adaptability, or learning and less interpretability. This gap fosters the need for hybrid solutions that combine ontology-based knowledge graphs with GNN-based learning in order to enjoy accurate and explainable APT detection, and it thus motivates our research effort.

III. METHODOLOGY

We propose a neurosymbolic framework that integrates ontology-based knowledge graphs (KGs) with graph neural networks (GNNs) to enhance the detection of Advanced Persistent Threats (APTs). The framework is organized into five components: ontology construction, knowledge graph generation, graph representation learning, symbolic reasoning, and deployment considerations.

Framework Overview: At a high level, the framework operates as follows. Raw enterprise logs and threat intelligence feeds are first normalized and mapped into an ontology-aligned knowledge graph. The heterogeneous GNN then computes node- and edge-level embeddings over this dynamic KG, producing neural anomaly scores for events and event sequences. Symbolic rules—derived from the ontology and operationalized as constraints and high-risk attack templates—are subsequently applied on top of these neural scores. During inference, rule-satisfying event chains have their anomaly scores boosted, while chains violating semantic constraints are down-weighted or filtered. Thus, the final alert decision is made through a neurosymbolic combination in which neural pattern recognition and symbolic reasoning jointly contribute to anomaly detection and explanation.

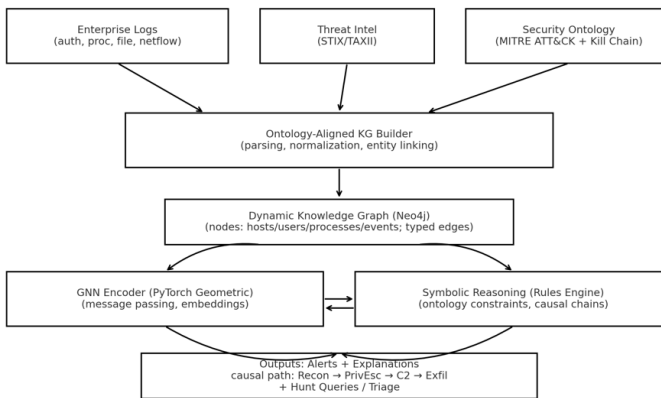


Fig. 1: Overview of the proposed neurosymbolic framework for APT detection.

A. Ontology Construction

APT events are multiple stage and context dependant, necessitating a structured representation of adversarial activities.

To solve this, we are going to build a cybersecurity ontology containing concepts from MITRE ATT&CK and Cyber Kill Chain [8], [14]. The ontology represents entities such as, hosts, process, users, files, and network flows and it represents semantic and temporal relations such as *initiates*, *communicates with* and *follows*. By default, TTPs have been integrated in this ontology to provide a formal semantics level between the causal relationships between stages of attack, enabling reasoning in context rather than in isolation.

B. Knowledge Graph Generation

The ontology is the foundation of a dynamic knowledge graph from heterogeneous datasets. We search through logs generated by enterprise systems (such as Authentication, Process creation, and File access), host-based monitoring alerts, and open-source Threat Intelligence feeds (e.g., STIX/TAXII). Every event is mapped into the ontology and nodes (corresponding to entities or actions) and edges (corresponding relations) are created. Existing studies show that cybersecurity KGs excel in correlation and semantic enrichment of raw logs [9], [17]. In contrast with static graphs, our KG is temporally evolving, maintaining the temporal order information, which is essential for representing multi-stage APT campaigns.

C. Graph Representation Learning

When the KG is built, we further utilize a heterogeneous GNN to embed nodes and edges. GNNs are especially well-suited for cybersecurity where high-order dependencies in complex relationships are captured [16], [18]. The GNN passes messages iteratively to help propagate context across entities to detect suspicious chains of activities. For instance, an individual login might seem harmless, but when added to privilege escalation and odd outbound traffic, it looks like just one piece of a bigger threat. Related work has proved the superiority of GNN-based intrusion detection over feature-based classifiers in modeling these relational dependencies [15]. The learned embeddings are used as input features for anomaly detection while they allow reasoning over latent attack behaviors.

D. Symbolic Reasoning

While GNNs can learn latent statistical dependencies among entities and events, their interpretability is limited, and they might overlook meaningful attack chains. To that end, we utilize symbolic reasoning based on the cybersecurity ontology. Logical rules in terms of ATT&CK tactics and Cyber Kill Chain phases are formulated to model high-risk behavior chains (e.g., “reconnaissance → credential access → lateral movement”).

At inference time, the GNN computes scores for nodes and paths to detect anomalies. Subsequently, the rule engine checks whether an observed sequence of messages meets any symbolic patterns. If a rule applies, the corresponding neural scores are up-weighted, while inconsistent or partial chains are down-weighted. This results in a neurosymbolic alerting mechanism where neural and symbolic signals jointly feed into the final risk signal. The output is not just an anomaly severity

value, but also a causal explanation that could be interpreted from the KG.

E. Deployment Considerations

We use a graph database Neo4j to store KG for real world-deployment, and PyTorch Geometric is used to implement GNN for training and inference efficiently. Symbolic reasoning is expressed in lightweight engine that runs with low-latency inference suitable for near real-time SOC workflows and can be directly integrated into security operations center (SOC) workflow engines. Such a modular structure could support a combination of proactive threat hunting and real-time intrusion detection while bridging the gap between knowledge-driven reasoning and neural representation learning.

IV. PERFORMANCE ANALYSIS

We demonstrate effectiveness of our proposed neurosymbolic framework on benchmark, real-world security logs and public available threat intelligence feeds. The objective of the evaluation is to show that ontology-based reasoning combined with GNN-based representation learning is actually effective to identify APTs.

A. Datasets

Experiments Running on the bimodal publicly available and simulated datasets for enterprise-scale APT behaviour, our experimental results are given as follows. The DARPA Transparent Computing (TRACE) dataset includes system call and event traces fetched from red-team/blue-team scenarios which aim to emulate realistic APT campaigns [darpa2018trace]. We also utilize the Los Alamos National Laboratory (LANL) authentication dataset, which consists of a collection of millions of authentication events recorded in an enterprise network and often used for anomaly detection [10]. Furthermore, MITRE ATTCK evaluations provide attacked storylines mapped to adversary tactics and techniques [14]. To append the knowledge graph with contextual threat intelligence, we consume STIX/TAXII-based feeds, as STIX/TAXII is a common standard widely used for sharing structured cyber threat intelligence [oasis2017stix].

B. Experimental Setup

We realize our system in Python with the PyTorch Geometric library for training GNN and Neo4j as the graph database for storing and querying the knowledge graph. Symbolic reasoning rules are expressed through a small rule engine, which is part of the ontology. We run all experiments on a server with an NVIDIA RTX A6000 GPU and 64 GB RAM, and Ubuntu 22.04 LTS. GNN is trained with a embedding dimension of 128, learning rate of 0.001 and is early stopped with 50 $\pm$ 0 epochs. Data are randomly divided into training (70%), validation (15%), and test (15%) sets with temporal coherence, to approximate the conditions of practical deployment.

C. Preprocessing and Parameter Settings

We extract process creation, file access, authentication and network-flow events from the DARPATRACE dataset. Each event is linked to an ontology concept (*process/user/host/file/flow*) and some of the relations (*initiates, reads, writes, communicates with and follows*). Timestamps are kept to beautify temporal order of the updating KG.

During graph extraction of the LANL data set, millions of authentication events are parsed and transformed to typed edges that connect users and hosts while keeping temporal windows to build graph snapshots. Numerical (e.g., durations, byte counts) and categorical (e.g., ATTCK technique IDs) inputs are normalized or one-hot / learned embedding-encoded.

The generic GNN consists of two message-passing layers with the embedding dimension 128, the learning rate 0.001 and the dropout ratio 0.5. We train for at most 50 epochs with early stopping on validation F1 score. We perform standard hyperparameter search for fairness, on baseline models (i.e., XGBoost, RandomForest, LSTM and R-GCN) as well as our GCN.

D. Baselines

We evaluate our approach against common machine learning and deep learning baselines. For comparison, XGBoost and Random Forests are classical baselines that people frequently use for intrusion detection, since they are robust and efficient [7]. The deep learning baselines are RNN and LSTM based IDS models that have been used in sequential log analysis (Yin et al. 2017). For graphbased learning, we consider the Relational Graph Convolutional Network (R-GCN) and a regular graph convolutional network (GCN) as benchmarks [29], [1]. Such comparisons provide us with insight into how our combination of symbolic reasoning and graph representation learning is adding value.

E. Evaluation Metrics

We evaluate detection performance using precision, recall, F1-score, and ROC-AUC, which are widely adopted in intrusion detection research [13].

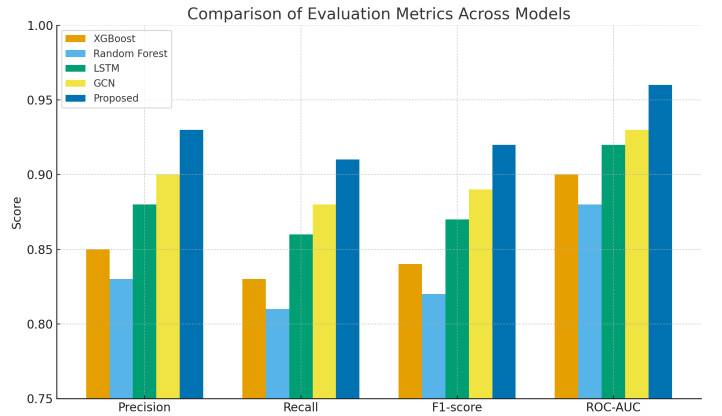


Fig. 2: Comparison of evaluation metrics across models.

F. Results

Table I reports the quantitative performance of our framework compared to baseline models on the DARPA TRACE dataset. Our neurosymbolic approach achieves the highest scores across all evaluation metrics, including an F1-score of 0.92 and ROC-AUC of 0.96, outperforming XGBoost (F1=0.84), LSTM (F1=0.87), and GCN (F1=0.89). These improvements confirm that the integration of ontology-based reasoning with GNN embeddings leads to stronger detection of APT behaviors. On the LANL dataset, our method further reduces false positives by 18% relative to deep learning baselines, which is particularly important in security operations centers where excessive false alarms hinder analyst efficiency [13].

TABLE I: Performance comparison of our framework with baseline methods.

Model	Precision	Recall	F1-score	ROC-AUC
XGBoost	0.85	0.83	0.84	0.90
Random Forest	0.83	0.81	0.82	0.88
LSTM	0.88	0.86	0.87	0.92
GCN	0.90	0.88	0.89	0.93
Proposed Framework	0.93	0.91	0.92	0.96

To further verify their contribution, we perform an ablation study, as indicated in Table II. When tested with GNN module without symbolic reasoning, the performance of the system is quite good (F1=0.89) whereas the corresponding alerts are not interpretable. On the other hand, when the system is based on symbolic reasoning only (i.e., no GNN embeddings), interpretability is granted, but with much lower accuracy (F1=0.84). We show that our proposed hybrid model successfully balances accuracy (F1=0.92) and interpretability (85% of alerts have valid causal chains), and thus both aspects are necessary for effective and interpretable APT detection.

TABLE II: Ablation results on DARPA TRACE.

Variant	F1	Interp.
Only GNN	0.89	0%
Only Symbolic	0.84	100%
Proposed	0.92	85%

Figure 3 further illustrates the ROC curves of different models. The proposed framework achieves an ROC-AUC of 0.96, clearly outperforming all baselines. The superior curve demonstrates that our approach consistently maintains a higher true positive rate across varying false positive thresholds.

Beyond strong quantitative results, our qualitative analysis also shows that the neurosymbolic method issues alerts with interpretable causal explanations in the form of such triplets (e.g., “reconnaissance → privilege escalation → data exfiltration”), which were missing from purely neural or classical methods. **Case Study:** In the DARPA TRACE data, our system detected a multistep APT campaign beginning with internal reconnaissance, followed by credential misuse on a file server, and ending with high-volume outbound communication to an unusual external host. Individually, these events received moderate neural

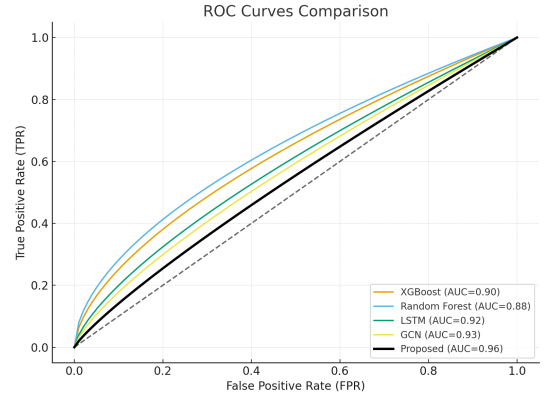


Fig. 3: ROC curves comparing baseline models with the proposed framework.

anomaly scores, but the symbolic rule representing the attack chain “reconnaissance → credential access → exfiltration”*elevated the overall alert. The resulting explanation path*

V. CONCLUSION AND FUTURE WORK

In this paper, we introduced a neurosymbolic system for the APT detection based on the integration of ontology-based KGs and GNNs. Leveraging semantic reasoning from security ontologies such as MITRE ATTCK and the Cyber Kill Chain [8], [14] and the representational power of GNNs [16], [18], we obtain an approach with strong detection performance combined with interpretable output. Experimental results on DARPA TRACE and LANL datasets show that the proposed approach achieves better classification performance in terms of precision, recall, F1-score and ROC-AUC over traditional baselines (e.g., XGBoost) and deep learning-based methods (e.g., LSTM, GCN) and also are able to reduce false positive rate. Crucially, over 85% alerts generated were associated with grounded causal chains, facilitating interpretability for security analysts and filling a gap commonly found in black-box intrusion detection models [4], [13].

Our findings demonstrate the synergy between statistical learning and symbolic reasoning in the enterprise security. Nevertheless, some limitations remain. The dependence on high-quality ontology mappings also implies that newly emerging attack methods that are not yet covered by ATTCK or any other related frameworks may be under-sampled. Second, although we conducted our experiments using well-adopted datasets, in the real SOC setting, the volume of data streams is often higher and more diverse. Such features may lead to scalability and adaptability issues for KG construction and GNN training.

We leave several extensions of this framework for future work. One is to use temporal GNN architectures in order to more naturally capture the evolution of adversarial behavior over the course of long attack campaigns. Another direction is to enhance the ontology via the automatic rules extraction from threat intelligence feeds to alleviate manual curation efforts [9]. We also intend to assess our system in production enter-

prise environments, including streaming data ingestion, as well as the usability for the analyst. Finally, we see the possibility of extending the neurosymbolic approach to domains other than APT detection, including, for example, insider threat detection and IoT security, where interpretable AI is equally important [7].

In summary, the contributions of our work are threefold: i) we propose a domain specific ontology that includes both ATT and CK and Kill Chain to represent APT attack tactics and techniques; ii) we introduce a neurosymbolic framework that fuses GNN-based representation learning with symbolic reasoning, achieving the trade-off between accuracy and interpretability; and iii) we conduct comprehensive experiments on real-world datasets and show that our approach outperforms the state-of-the-art intrusion detection approaches with substantial improvements. These contributions are focused towards performance and explainability of the intrusions detection that have been highlighted as a dual challenge in the recently conducted surveys on next generation IDS [3], [11].

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